RAMCO INSTITUTE OF TECHNOLOGY

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Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year: 2023 - 2024 (Odd Semester)

Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC3353 & Electronic Devices & Circuits
Semester, Degree & Branch : III B.E ECE B
Topic : Full Wave Rectifier
Innovative Method : One Minute Paper

05/10/23 R. Sowmiya 953622106093 1) rectifier → ~~Converter~~ Convert AC to pulsating DC 2) Repel Ripple factor factor for fullwave → ~~0.121~~ 81.3% 3) V_{RMS} of full wave ⇒ $\frac{2V_m}{\pi}$ 4) Efficiency for full wave ⇒ ~~81~~% 5) capacitor filter	Reg.No 953622106065 Date: 05/10/2023. 1. Rectifier is convert ~~DC~~ to AC. 2. Ripple factor: 0.8106% (or) 8.2%. 3. Vrms = ~~Vm~~ $2V_m/\pi$ 4. Efficiency = 0.8106 5. With filter → Capacitor without filter

Signature of the faculty member

HOD

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Department of Electronics and Communication Engineering
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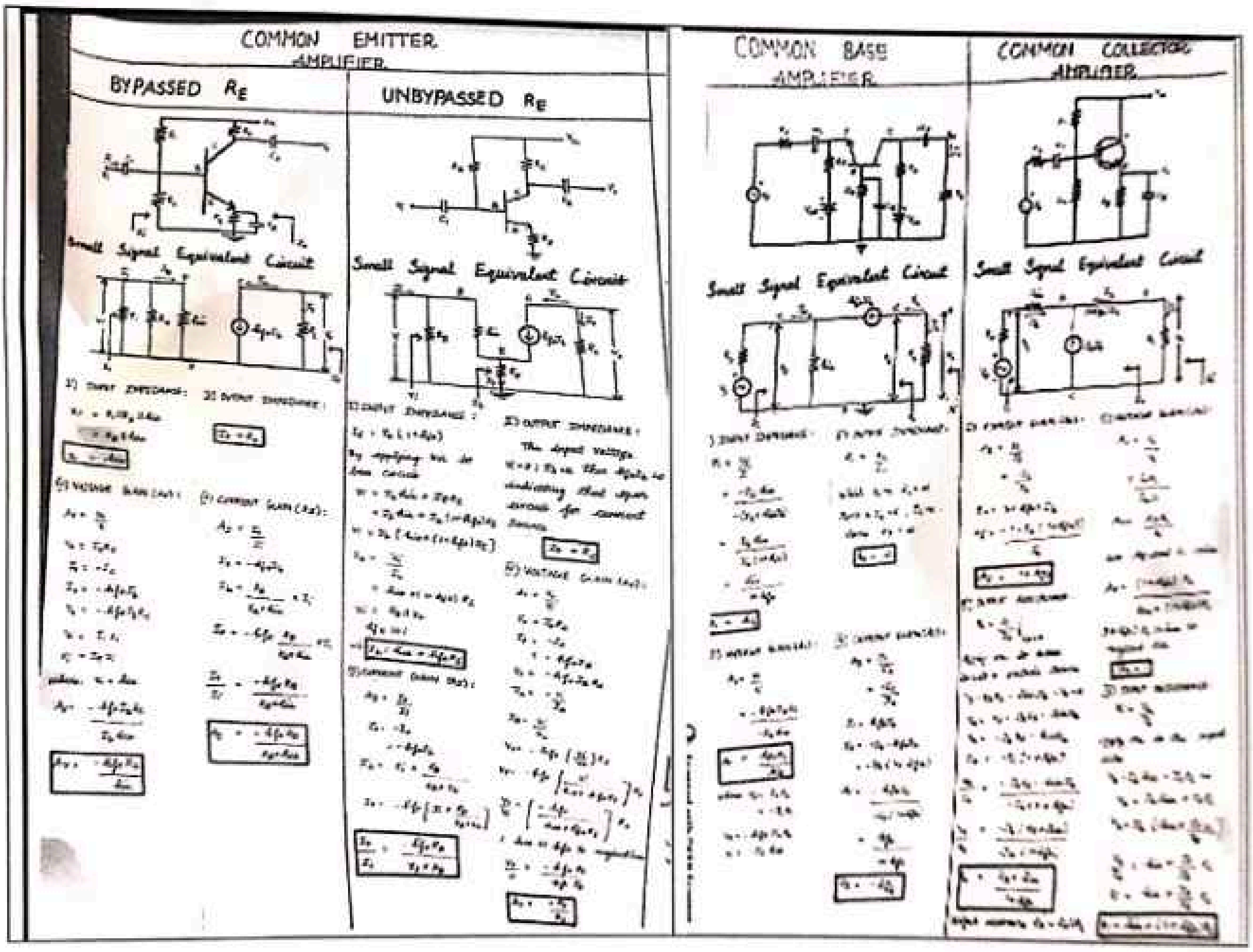
Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : EC3353 & Electronic Devices & Circuits
Semester, Degree & Branch : III B.E ECE B
Topic : Small Signal Analysis of BJT
Innovative Method : Chart Preparation



12/10/23
Signature of the faculty member

HOD

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)
Degree, Semester & Branch: III Semester B.E. ECE B
Course Code & Title: EC3353 Electronic Devices and Circuits
Name of the Faculty member (s): Mrs.V.SrirengaNachiyar

Innovative Practice Description

- **Unit / Topic: Unit V** / DC/DC convertors –Buck-Boost analysis and design
- **Course Outcome:** CO 5
- **Unit Outcome:** 19
- **Activity Chosen: Flipped class**
- **Justification:**

The flipped classroom model is based on the idea that traditional teaching is inverted in the sense that what is normally done in class is flipped or switched with that which is normally done by the students out of class. This Activity gives the clear understanding of the particular topic.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

Students were given the task of preparing a presentation on the topic of " DC/DC convertors –Buck-Boost analysis and design". They described about the DC converters and important component present in DC converters". Students were divided into groups and each group will prepared the detailed presentation of DC converters and its types and deliver their presentation among all the students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO3
CO5	2	1	2	1	2	2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10	PO12
	2	1
Justification for correlation	Students will understand the devices used in DC Converters to convert AC into DC and will give clear information during discussion session. Hence it is mapped at level 2	Students were identify the previous instances in DC Converter system and how it can be used where improvements in technology required lifelong learning for practitioners in accordance with changing. Thus level 1 is mapped

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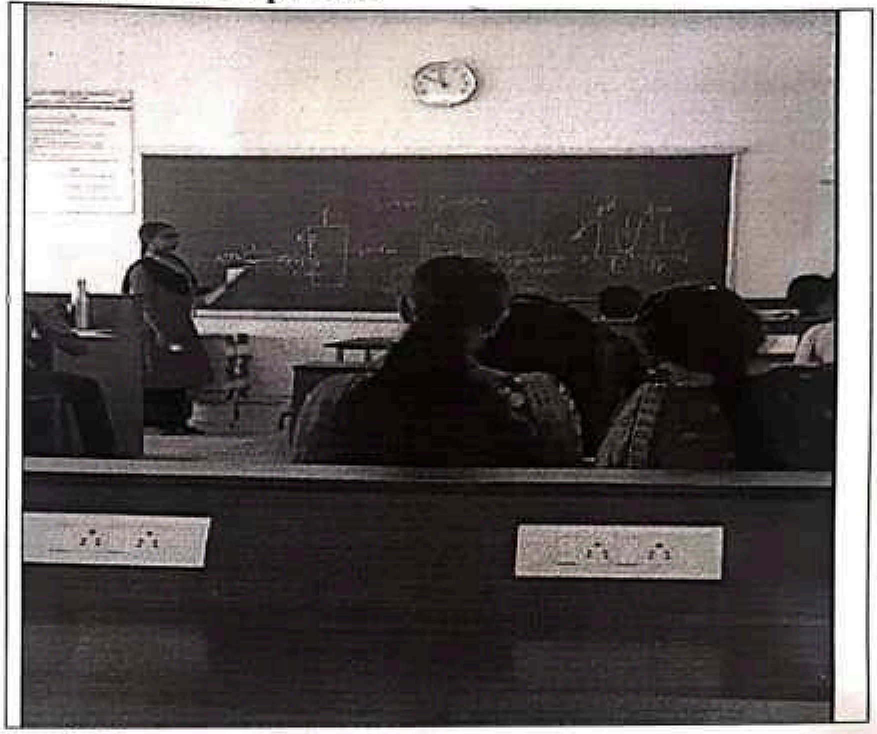
- **Images / Screenshot of the practice:**



Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***

  The practise was well-received by students, who said it helped them remember subjects and substance while writing exams. They also noted that because they are presenting in front of all of the students, this activity boosts their confidence.

- ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

  The Students eagerly participated in that activity and then they acquired the knowledge of various DC Converters. Due to this activity, the students were able to recall the topic and remember the important terms easily.

- ***Challenges faced in implementation:***

  I have planned this activity for 30 minutes. Students take some time to present the topic, they present the topic with presentation. Students asked doubts regarding the topic before presentation. I've given a brief overview of the DC Converters used for sports and encouraged the students to present which it takes another 5 minutes for completing the activity.

References:

- David A. Bell, "Electronic Devices and Circuits", Oxford Higher Education press, 5 th Edition, 2010

21/12/23

Signature of the faculty member

HOD